

You can either specify the new image size in terms of the size in inches or centimetres, or you simply specify the new resolution. Note that an increase in resolution will increase the document size and vice versa (for the same dpi). When you only specify a new dpi, the resolution increases, but the document size does not change. After setting a new dpi, you can change the image size by altering the image resolution separately, or by changing the document size.

2.2.2 Resampling Algorithms

When increasing the resolution, pixels need to be created. A resampling algorithm (or logic) determines the basis on which the new pixels are calculated and used. Generally, a new pixel is introduced between existing pixels by performing some calculation on the existing ones in order to determine (or predict) the value of the new pixel. Photoshop provides you with several resampling options: Nearest Neighbour, Bilinear, Bicubic, Bicubic Smoother and Bicubic Sharpen. Of these, the most efficient is the Bicubic algorithm, which looks at the adjacent pixels and computes the value of the pixel that needs to be added.

The Nearest Neighbour method is best avoided in most cases—it simply copies the value of the nearest pixel onto the new pixel. Bilinear is somewhat better—it averages the values of the surrounding pixels rather than simply copying them. But it is still not as effective as the Bicubic method. In Photoshop CS, two new algorithms were introduced—Bicubic Smoother and Bicubic Sharpen. You can try these and see what works best for your image. However, smoothening and sharpening can be done by separate filters as well.